

TYRMA, Russian Federation

WMO: 315380

Lat: 50.0456N

Lon: 132.1770E

Elev: 347

StdP: 97.22

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.0	-34.8	-39.7	0.1	-36.7	-37.5	0.1	-34.5	4.3	-21.0	3.7	-20.8	0.4	30	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	29.1	20.2	27.3	19.2	25.7	18.3	21.7	26.1	20.7	25.1	19.9	24.0	2.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.6	23.8	19.3	14.6	22.8	18.5	13.9	21.9	65.4	26.2	61.9	25.1	58.8	24.1	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
5.5	4.7	4.0		-39.3	32.7	1.9	1.7	-40.7	33.9	-41.8	34.9	-42.9	35.9	-44.3	37.1
			DB	-39.4	23.3	1.9	1.4	-40.8	24.3	-41.9	25.1	-42.9	25.9	-44.3	26.8
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-25.3	-18.9	-8.1	2.4	11.1	16.8	20.0	18.0	11.0	1.3	-12.6	-23.9
	DBStd	16.64	4.64	5.20	6.09	4.73	3.88	3.47	2.57	2.90	4.02	4.71	7.01	5.43
	HDD10.0	4768	1095	810	561	233	33	1	0	1	35	271	678	1052
	HDD18.3	7029	1353	1043	819	478	227	69	13	40	221	528	928	1310
	CDD10.0	903	0	0	0	5	66	205	309	250	66	2	0	0
	CDD18.3	121	0	0	0	0	1	24	64	31	1	0	0	0
	CDH23.3	1027	0	0	0	5	62	280	449	216	14	0	0	0
	CDH26.7	231	0	0	0	0	11	76	113	29	1	0	0	0
(14) Wind	WSAvg	1.5	0.9	1.3	1.7	2.0	2.0	1.5	1.4	1.2	1.4	1.7	1.5	1.0
(15) Precipitation	PrecAvg	659	7	8	13	34	70	95	131	131	88	46	19	13
	PrecMax	843	24	31	30	71	142	187	224	206	162	111	44	32
	PrecMin	482	0	0	1	4	23	23	13	48	19	14	2	3
	PrecStd	97	6	7	9	19	32	46	49	46	37	22	12	8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-9.6	-2.8	9.2	23.0	28.0	31.7	33.0	29.0	25.2	18.1	5.0	-7.6
		MCWB	-10.6	-5.6	2.8	10.7	14.8	19.6	20.7	21.3	17.3	10.0	1.1	-8.7
	2%	DB	-13.7	-5.0	5.8	17.5	24.8	28.7	29.5	27.4	22.7	13.4	2.0	-10.5
		MCWB	-14.4	-7.1	0.5	8.2	12.9	19.1	20.7	20.4	15.8	7.0	-0.1	-11.1
	5%	DB	-16.2	-7.3	3.5	13.6	22.2	26.5	27.6	25.7	20.3	10.4	0.0	-12.8
		MCWB	-16.8	-8.8	-0.6	6.0	11.7	17.7	20.1	19.3	14.4	5.5	-1.7	-13.4
	10%	DB	-18.2	-9.9	1.4	10.9	19.8	24.4	25.8	24.2	18.0	8.5	-1.9	-15.8
		MCWB	-18.8	-11.2	-1.9	4.4	10.9	16.7	19.5	18.4	13.3	4.6	-3.3	-16.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.6	-5.2	3.0	11.9	15.9	21.8	23.0	23.2	18.6	10.6	2.0	-8.5
		MCDB	-9.5	-3.1	8.0	20.3	25.8	28.0	28.2	27.2	23.0	16.5	3.7	-7.6
	2%	WB	-14.5	-6.9	1.2	9.0	13.9	19.7	21.8	21.6	16.9	8.2	0.0	-11.0
		MCDB	-13.8	-5.0	4.9	16.7	22.1	26.4	26.5	25.4	20.8	11.9	1.7	-10.5
	5%	WB	-16.8	-8.8	-0.3	6.7	12.8	18.7	21.0	20.6	15.6	6.4	-1.5	-13.4
		MCDB	-16.2	-7.3	3.1	13.0	19.9	24.6	25.6	24.1	19.0	9.5	0.0	-12.8
	10%	WB	-18.7	-11.2	-1.7	4.8	11.9	17.7	20.2	19.5	14.2	4.8	-3.4	-16.3
		MCDB	-18.2	-10.0	1.3	10.4	18.0	23.0	24.7	22.6	17.3	8.0	-1.9	-15.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	13.7	12.6	11.5	13.0	11.7	10.3	10.9	12.0	10.6	10.3	9.9
		MCDBR	11.3	13.5	13.0	17.3	18.8	16.3	13.8	13.8	15.9	14.5	10.1	9.0
		MCWBR	10.5	11.7	8.6	9.4	8.3	6.8	5.9	6.7	8.5	8.7	7.8	8.4
	5% WB	MCDBR	11.3	13.3	11.9	16.0	15.6	14.2	11.2	10.8	12.4	12.7	9.5	9.0
		MCWBR	10.5	11.5	8.0	8.9	7.3	6.3	5.5	5.9	7.1	8.1	7.5	8.5
(40) Clear-Sky Solar Irradiance	taub		0.268	0.302	0.372	0.500	0.557	0.502	0.484	0.422	0.389	0.399	0.332	0.274
	taud		2.160	2.058	1.919	1.777	1.767	2.001	2.109	2.284	2.288	2.075	2.088	2.117
	Ebn at Noon		759	815	808	738	717	767	776	811	797	686	647	678
	Edh at Noon		76	111	154	200	212	169	149	120	107	110	83	69
(44) All-Sky Solar Radiation	RadAvg		1.72	2.91	4.41	4.85	4.99	5.46	5.04	4.54	3.60	2.36	1.61	1.33
	RadStd		0.04	0.07	0.24	0.51	0.48	0.71	0.43	0.38	0.37	0.18	0.12	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page